Coefficient height one and an eleven-integer degree-464 equation for Con(ZFC)

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Abstract

We give two explicit ordinary polynomial equations over unrestricted integers whose lack of solutions is provably equivalent in ZFC to syntactic consistency of ZFC, using the explicitly retained calibrations of the ProveIt repository. The first is a fully expanded quartic with 7,431 variables and 34,086 nonzero coefficients, every one equal to 1 or -1. Its constant coefficient is one, and its individual variable degrees are at most two. A general block-and-cycle construction gives an exact bijection between its zero set and that of a supplied direct C222 quadratic system. This route does not simulate K215 and does not instantiate Jones's universal polynomial.

The second equation has eleven variables, exact total degree 464, and a division-free circuit of 122 live binary arithmetic operations. It combines selective coprime radical fusion with powered divisibility, fixed-pair Pell interpolation, a reduced guard, and a shared auxiliary center. A direct companion has degree 488 in 121 operations. The eleven-variable construction retains the complete fixed native coefficient tuple. Both reductions have ordinary mathematical proofs; their finite certificates, source bytes, exact polynomial data, and reproducible checkers are supplied. No global simplicity minimum, worldwide priority, proof of consistency, or new end-to-end formal proof-assistant certificate is asserted.

New equation	Integer variables	Exact degree	Principal size objective
Height-one quartic $\mathcal{H}$	7,431	4	$\mathrm{ht}(\mathcal{H}) = 1$
Centered norm $\mathcal{E}_{464}$	11	464	122 arithmetic operations
Direct norm $\mathcal{E}_{488}$	11	488	121 arithmetic operations

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1 Objective, comparison, and logical contract

The requested target is a fixed polynomial $P \in \mathbb{Z}[x_1, \ldots, x_n]$ such that

$$\mathsf{ZFC} \vdash (\forall x_1, \ldots, x_n \in \mathbb{Z}\ P(x_1, \ldots, x_n) \neq 0) \longleftrightarrow \mathrm{Con}(\mathsf{ZFC}). \tag{1}$$

Here $\mathrm{Con}(\mathsf{ZFC})$ means the absence of a finite syntactic proof of contradiction in the fixed standard proof system. Every variable in this report is unrestricted unless explicitly used in an intermediate lemma. Positive, nonnegative, and unrestricted variable counts are not interchanged.

The classical Diophantine representation theorem establishes the availability of such equations in principle. Recent work on universal pairs and reductions over the integers refines the number of unknowns and degree [5, 6]. Those general results do not select the smallest description for this particular consistency statement. A source-specific construction must also fix its coefficient tuple or literal residual list and establish its refutation calibration.

There is no single agreed meaning of "simplest." We measure the vector

(domain, variables, degree, coefficient height, support, circuit operations, coefficient storage).

The coefficient height used throughout is $\mathrm{ht}(P) = \max_\alpha |c_\alpha|$ for the *collected* polynomial $P = \sum_\alpha c_\alpha x^\alpha$. It is not the weighted sum sometimes also called a height in experimental Diophantine searches. A circuit operation is a live binary addition, subtraction, or multiplication, including squares and multiplication by fixed constants; leaves and reuse are free. This count does not measure bit complexity.

1.1 Frozen sources

The comparison snapshot is the exact ProveIt commit

`45b399ec51dfbecb891dd51d8a9e4b6b7c361ac6`.

The audit included the active research summaries, the delivered forms through package 110, and all eight then-present incoming archives. The incoming archives matter: their results already improve on several numbers in the collection README. In particular, the direct source chosen here is the 519-variable equation in the powered-saturation archive, while the eleven-variable comparison comes from the coprime projection archive. Their immutable paths and SHA-256 hashes are retained with the artifacts.

Compared construction	Variables	Degree	Additional metric
Incoming height-two quartic [4]	3,564	4	Height 2; 14,518 terms
Incoming direct source [2]	519	4	Height 411,060; 6,253 terms
New height-one quartic	7,431	4	Height 1; 34,086 terms
Incoming centered coprime norm [3]	11	688	126 operations
Incoming direct coprime norm [3]	11	728	122 operations
New centered norm	11	464	122 operations
New direct norm	11	488	121 operations

All seven rows use unrestricted integer coordinates. The height-one construction improves the coefficient objective while increasing dimension and support. The centered norm lowers degree by $224/688 \approx 32.56\%$ and lowers operation count from 126 to 122 relative to its matched centered source. Ten-integer equations and a native 37-integer quartic already occur in the inspected repository. Neither new equation is advertised as the smallest by variable count.

1.2 Three kinds of evidence

The report distinguishes ordinary unbounded proofs, exact finite artifact checks, and formal proof-assistant endpoints. The new mathematics below is proved by ordinary arguments formalizable in ZFC. The scripts verify literal polynomial identities, coefficients, dependency graphs, degree attainments, and source bindings. These checks do not decide whether any equation has an integer zero. No new complete Lean, Rocq, or object-language ZFC proof has been built in this round. Inherited formal endpoints are named as part of the provenance rather than described as newly checked.

2 A quartic with coefficient height one

The principal new mechanism is a coefficient reduction that applies to any finite quadratic system over the integers. It uses nonnegative quadratic forms to enforce agreement between separate occurrences of a variable. The separation is essential: algebraically identifying these occurrences before collecting coefficients would destroy the height bound.

2.1 The block identity

Let $\sigma_1, \dots, \sigma_k \in \{-1, 1\}$. For two lists of real numbers $m = (m_1, \dots, m_k)$ and $w = (w_1, \dots, w_k)$, define

$$\mathcal{B}_\sigma(m, w) = \sum_{i=1}^{k} (m_i^2 + w_i^2 - m_i w_i) \tag{2}$$

$$+ \sum_{1 \le i < j \le k} \sigma_i \sigma_j (m_i m_j + w_i w_j). \tag{3}$$

The exact identity

$$2\mathcal{B}_\sigma(m, w) = \left(\sum_i \sigma_i m_i \right)^2 + \left(\sum_i \sigma_i w_i \right)^2 + \sum_i (m_i - w_i)^2 \tag{4}$$

proves both nonnegativity and the equivalence

$$\mathcal{B}_\sigma(m, w) = 0 \quad \Longleftrightarrow \quad w_i = m_i \ (1 \le i \le k), \qquad \sum_i \sigma_i m_i = 0. \tag{5}$$

These statements hold over any ordered field, and consequently over $\mathbb{Z}$ as well as $\mathbb{R}$. The factor $1/2$ in the proof is harmless: the definition (3) is already an integer polynomial.

Ordinary squaring of a residual produces coefficients 2 in its mixed terms. Formula (3) distributes each diagonal contribution between two matching constraints. After division by two, every displayed coefficient is 1 or -1. The remaining issue is to prevent distinct displayed terms from becoming the same monomial.

2.2 Separating occurrences and preserving degree four

First write each source row as a list of signed monomial occurrences,

$$R_j(x) = \sum_{i=1}^{k_j} \sigma_{ji} M_{ji}(x), \qquad \deg M_{ji} \le 2. \tag{6}$$

A coefficient c can be written as $|c|$ occurrences with sign $\operatorname{sgn}(c)$. Section 2.5 gives a more economical preliminary normalization for large coefficients. Occurrences remain distinct even when their monomials agree in the source row.

For each occurrence introduce two entirely fresh variables a_{ji}, b_{ji} and set $m_{ji} = a_{ji}b_{ji}$. Label the two factors as follows:

Source monomial	Label of a_{ji}	Label of b_{ji}
$x_p x_q$	x_p	x_q
x_p	x_p	1
1	1	1

Even for x_p^2, the two aliases are different variables with the same label. No alias is used in two occurrences. Introduce one further fresh variable w_{ji} for each occurrence, and use the block $\mathcal{B}_{\sigma_j}(m_j, w_j)$.

All m_{ji} have degree exactly two, including padded linear and constant occurrences. Thus the $m_{ji}m_{j\ell}$ terms have degree four, the $m_{ji}w_{ji}$ terms have degree three, and the $w_{ji}w_{j\ell}$ terms have degree two. This deliberate separation of degrees is what allows the agreement constraints below to be added without coefficient collisions.

2.3 Agreement cycles and the single unit anchor

For a list $z_1, \dots, z_d$ with $d \geq 3$, put $z_{d+1} = z_1$ and define

$$\mathcal{C}(z_1, \dots, z_d) = \sum_{i=1}^{d} z_i^2 - \sum_{i=1}^{d} z_i z_{i+1} = \frac{1}{2}\sum_{i=1}^{d}(z_i - z_{i+1})^2. \tag{7}$$

This polynomial is nonnegative and vanishes exactly when all its entries agree. Its coefficients are 1 and -1. The restriction $d \geq 3$ is necessary: at $d = 2$ the two edge monomials coincide and give coefficient -2.

Group all aliases carrying the same old-variable label x_p. For a group of size at least three, add its cycle (7). For a group of size two, add one fresh dummy alias with the same label and use the resulting triangle. A singleton needs no agreement constraint. An unused source variable can be omitted when only existence of solutions matters; all source variables of the instantiated equation occur.

List all aliases with label 1 as $u_1, \dots, u_d$. Add fresh unit aliases if necessary so that $d \geq 2$, and put

$$\begin{aligned} \mathcal{U}(u_1, \dots, u_d) &= 1 + \sum_{i=1}^{d} u_i^2 - u_1 - u_d - \sum_{i=1}^{d-1} u_i u_{i+1} \\ &= \frac{1}{2}\left[(1-u_1)^2 + \sum_{i=1}^{d-1}(u_i - u_{i+1})^2 + (u_d - 1)^2\right]. \end{aligned} \tag{8}$$

It vanishes exactly when every $u_i = 1$. Only this block contributes a constant term, namely 1, or linear terms, namely $-u_1 - u_d$.

Definition 2.1 (The height-one polynomial)**.** For the fixed occurrence lists (6), define

$$\mathcal{H} = \sum_j \mathcal{B}_{\sigma_j}(m_j, w_j) + \sum_{\text{old-variable groups}} \mathcal{C} + \mathcal{U}. \tag{9}$$

The second sum includes the prescribed triangles for groups of size two and omits singletons. All sums are fixed and finite. All coordinates range independently over $\mathbb{Z}$.

Theorem 2.2 (Coefficient and zero-set theorem)**.** *For a finite quadratic integer system with at least one nonzero monomial occurrence, the construction above produces an ordinary polynomial $\mathcal{H}$ with the following properties:*

1. *Every nonzero coefficient of its fully collected expansion is* 1 *or* -1*; its constant coefficient is* 1*.*
2. *Its total degree is exactly four, and its degree in each individual variable is at most two.*
3. *It is nonnegative on every real tuple.*
4. *It has an integer zero if and only if all source rows have a simultaneous integer zero.*
5. *If every source variable occurs, projection to one representative of each old-variable alias group is a bijection on integer zero sets. Its inverse is explicitly polynomial.*

Proof. Nonnegativity follows from (4), (7), and (8). Suppose that $\mathcal{H} = 0$. Every block is nonnegative, so every block is zero. The cycles make each old-variable alias group constant; choose its common value as x_p. The anchor makes all unit aliases equal to one. Consequently each m_{ji} becomes its source monomial $M_{ji}(x)$. Formula (5) then gives $R_j(x) = 0$ for every source row.

Conversely, given a simultaneous source zero x, set every old-variable alias to its label's value, every unit alias to one, every agreement dummy to its group's value, and $w_{ji} = M_{ji}(x)$. Every block vanishes. All added values are uniquely forced on a zero, which proves the bijection and its explicit inverse. Before coefficient normalization, this inverse has degree at most two in the source variables.

For the coefficient claim, partition the collected monomials by degree and by the variable families they contain. At degree four, the terms m_{ji}^2 use exactly one private alias pair, each with exponent two; the terms $m_{ji}m_{j\ell}$ use two distinct private pairs. Each pair belongs to precisely one occurrence, so neither kind can collide with another term. Across different rows all alias supports are disjoint.

At degree three, each $-m_{ji}w_{ji}$ contains its own private w variable and its private alias pair. These monomials are all different. At degree two, the terms containing w variables come only from the row blocks. A square uses one w index; a mixed term uses a pair from the same row. They are distinct, and no w is an alias. The remaining quadratic terms are precisely the agreement cycles and unit anchor. Their alias groups are disjoint. Every cycle has at least three distinct vertices, so each undirected edge occurs once. The anchor path likewise has no repeated edge. Its two linear terms are distinct because $d \geq 2$. The constant term occurs once. Thus every displayed term contributes to a distinct monomial, proving height exactly one.

Any occurrence contributes $a_{ji}^2 b_{ji}^2$ with coefficient one, so degree four is attained. The same enumeration shows that no individual exponent exceeds two. □

The polynomial is a rational sum of squares, through the identity for $2\mathcal{H}$. An integer sum-of-squares representation of $\mathcal{H}$ itself is neither used nor asserted. The proof relies on the order of $\mathbb{Z}$ or $\mathbb{R}$; reducing the single equation modulo a prime does not preserve its zero-set interpretation.

2.4 A small example: one addition equation

For $x + y - z = 0$, take three unit aliases u_1, u_2, u_3, put $m_1 = xu_1$, $m_2 = yu_2$, $m_3 = zu_3$, and introduce w_1, w_2, w_3. Then

$$\begin{aligned}\mathcal{H}_+ = \sum_{i=1}^{3}(m_i^2 + w_i^2 - m_i w_i) + m_1m_2 - m_1m_3 - m_2m_3 \\ + w_1w_2 - w_1w_3 - w_2w_3 + 1 + u_1^2 + u_2^2 + u_3^2 - u_1 - u_3 - u_1u_2 - u_2u_3.\end{aligned}$$

Its collected coefficients are all ± 1, its degree is four, and $\mathcal{H}_+ = 0$ holds exactly when

$$x + y = z, \qquad (u_1, u_2, u_3) = (1, 1, 1), \qquad (w_1, w_2, w_3) = (x, y, z).$$

This example shows why addition is compatible with height one. No positivity restriction on x, y, z, or on any auxiliary variable, is needed.

2.5 Normalizing large coefficients

Expanding a coefficient of magnitude c into c occurrences is correct but can be wasteful. The delivered compiler first constructs each needed positive integer c from a shared unit and smaller constants. For $c = 2d + \epsilon$ with $\epsilon \in \{0, 1\}$, introduce a constant wire C_c and the row

$$C_c - C_d - C_d - \epsilon = 0. \tag{10}$$

The repeated C_d is retained as two signed occurrences. Constant wires are ordered by their defining dependencies. Here C_1 denotes the literal constant 1 and requires no additional coordinate. The resulting finite triangular system forces each C_c to equal its stated integer and has a unique extension of every source tuple. Sharing the binary ancestors avoids repeating the same constant construction.

For a coefficient multiplying a quadratic monomial $x_p x_q$, introduce or reuse a scaled wire $S_{c,p}$ with

$$S_{c,p} - C_c x_p = 0, \tag{11}$$

and replace cx_px_q by $S_{c,p}x_q$. A linear term cx_p may be written directly as C_cx_p, and a constant c as C_c. Signs remain in σ_{ji}. All new rows are quadratic and use only unit signed occurrences. Substituting the forced constants and scaled wires restores each original source row exactly. This is a polynomial identity, independently checked in the release, rather than a test on sampled source zeros.

The binary normalization and the height-one construction compose. Every source tuple has a uniquely specified extension to the new coordinates at a zero. For the delivered normalization, its constant wires are constant and its scaled wires are linear in the source variables; the full inverse on zero sets therefore still has degree at most two. In fact, if L denotes this explicit extension, the stronger identity

$$\mathcal{H}(L(x)) = \sum_j R_j(x)^2 \tag{12}$$

holds for every source tuple, not only for source zeros. The constant and scaling rows vanish identically under L, each pair of row squares in (4) contributes twice the original squared residual, and all agreement and unit blocks vanish.

2.6 Size accounting and the instantiated equation

Let $L = \sum_j k_j$ be the number of signed occurrences after normalization. The row blocks use $3L$ variables: two aliases and one w per occurrence. Let d_2 count old-variable labels appearing

exactly twice among aliases, and let $\delta_u \in \{0, 1, 2\}$ be the number of additional unit aliases needed to make the unit group have size at least two. The exact count is

$$N = 3L + d_2 + \delta_u. \tag{13}$$

A block with k occurrences contributes $k^2 + 2k$ distinct monomials: k quartic squares, $\binom{k}{2}$ quartic cross terms, k cubic terms, and $k + \binom{k}{2}$ quadratic terms. A d-vertex agreement cycle contributes $2d$ monomials, and a d-unit anchor contributes $2d + 2$. These formulas are exact because the proof excluded collisions.

The instantiated source is the 519-integer, 321-row C222 system from the incoming powered-saturation packet [2], with its complete quadratic residual list included unchanged. The artifact inventory and measured counts appear in Section 5. In particular, the output contains every collected coefficient of $\mathcal{H}$; it does not leave an input program, compiler choice, or coefficient oracle to be supplied later.

The general construction changes the coefficient objective decisively: height one is the least positive coefficient height of any nonzero integer polynomial. This observation is only an optimum for that one coordinate of the complexity problem. It makes no claim that the resulting number of variables, support, or total description length is minimal. The existing height-two construction [4] has fewer variables, and the existing native quartic has far fewer variables with much larger coefficients. Those remain distinct tradeoffs.

2.7 Logical consequence and precise novelty scope

The construction is entirely finite. Its soundness and completeness use identities of integer polynomials, finite dependency induction, and nonnegativity of squares. These arguments are formalizable in ZFC. Composing Theorem 2.2 with the retained C222 calibration therefore gives

$$\mathsf{ZFC} \vdash \left[\forall z \in \mathbb{Z}^{7431},\ \mathcal{H}(z) \neq 0\right] \iff \mathrm{Con}(\mathsf{ZFC}). \tag{14}$$

The height-one transformation does not introduce another unbounded number-theoretic lemma. Its dependence on the inherited arithmetic encoding is identified separately in Section 4.

More generally, quadratic circuit conversion followed by this theorem reduces integer Diophantine solvability to multiquadratic quartics of height one and constant coefficient one. This is a direct corollary of the proved construction; it is not a claim to have newly established the classical undecidability theorem. Searches of the frozen repository, the current incoming packets, and selected literature did not locate this particular block-and-cycle compiler. That bounded audit is not a worldwide priority result. The concrete contribution is the proved compiler, its independently checked fully instantiated consistency polynomial, and its comparison with the supplied height-two endpoint.

3 Eleven integers at degree 464

This branch minimizes a different part of the complexity vector. It retains eleven coordinates and the complete native source34 coefficient tuple, but combines two incoming constructions in a way that substantially reduces degree. The three-root coprime projection of [3] supplies the variable count. Powered saturation and supported-modulus interpolation from [2] allow a much smaller scalar carrier. Two elementary changes to the guard and the radical weights then reduce the literal arithmetic cost.

3.1 The complete equation

The eleven independent integers are

$$j, C, z, h, r, v, w, T, c, n, s.$$

The six fixed coefficients are denoted by $\beta, K_0, M_C, M_F, A_0, B_0$. They are the unchanged source34 tuple, with

$$\beta = 2^{1064921} - 1, \qquad A_0 = 2^{1113}, \qquad B_0 = 2^{666104}.$$

The complete bytes of K_0, M_C, M_F, as well as the other three coefficients, are supplied with their bindings and finite reconstruction recipe. In particular, M_F already includes the source shift. None of these six coefficients is a free parameter or an existential unknown.

Define the following polynomial abbreviations:

$$\begin{aligned}
J &= 6j + 1, \qquad q = \beta J + 1, \qquad W = 12w + 4, \\
X &= qW, \qquad Y = (2v + 1)q^3, \qquad E = XY, \\
a &= E + Y, \qquad H = 4a + 3, \\
\Delta &= a^2 + H = (a + 2)^2 - 1, \qquad (15) \\
F &= (K_0 + W)C - z(q - 1), \\
R &= (q(q - F) - C)(q^2 - 1) + (M_C + qM_F)J, \\
k &= R + 1 + hE, \qquad \eta = c - Yk, \qquad (16) \\
D_{\mathrm{err}} &= (A_0F - q)^2 + (B_0C - q)^2, \\
\mathcal{B} &= k^2(q^2 - D_{\mathrm{err}}) - 4q^2\eta^2, \qquad D = 3\mathcal{B} - 4, \qquad (17) \\
L &= XY^2k, \qquad D_t = 1 + L(L + k), \\
D_g &= 1 + DD_ts^2, \qquad P = D_tD_g. \qquad (18)
\end{aligned}$$

The new strong and carrier scales, and the auxiliary data, are

$$\begin{aligned}
I &= 2rc^2, & S &= srH, & (19) \\
D_f &= 1 + \Delta I^2, & Q &= \Delta^2 I^2, & \delta &= Q - 1, \\
V &= D_f(c - R + 4IT) - c, \\
C_y &= QV^2 - 1, & D_a &= \delta C_y, & (20) \\
K &= X(3c - 2X) + 2, & Z_c &= HT + K, \\
A_c &= (Z_c^2 + 3H^2)(Z_c^3 - 19H^3). & & & (21)
\end{aligned}$$

For the centered equation set

$$Z = S(n + QV) - A_c, \qquad u = 4S^2D_f, \qquad b = S^2D_a, \qquad e = 16S^2P. \qquad (22)$$

Finally define

$$\boxed{\mathcal{E}_{464} = \left[(Z^2 - u - b - e)^2 - 4(ub + ue + be)\right]^2 - 64ubeZ^2 = 0.} \qquad (23)$$

Equations (15)–(23) specify a single ordinary integer polynomial. Abbreviations do not introduce coordinates, and all exponents are fixed nonnegative integers. The literal circuit contains 24 additions, 21 subtractions, and 77 multiplications: 122 live binary operations in total. The direct companion replaces Z by $Sn - A_c$ and has exact degree 488 in 121 operations.

The universal identity underlying (23) is

$$\mathcal{E} = \prod_{(\epsilon_1,\epsilon_2,\epsilon_3)\in\{-1,1\}^3} (Z - \epsilon_1\sqrt{u} - \epsilon_2\sqrt{b} - \epsilon_3\sqrt{e}). \qquad (24)$$

It holds algebraically even for zero, negative, or dependent radicands. One may prove it by successive quadratic conjugation, or work in the free algebra generated by three formal square roots. The checker verifies its entire polynomial expansion independently.

3.2 The inherited source interface

The old centered proof supplies four precise arithmetic interfaces, whose statements and full ordinary source proofs are retained in the package. They are used here in the order stated.

First, the disk inequality $\mathcal{B} \geq 2$ yields the common strict sign of $q, F, C, q - F - C$ and

$$k \neq 0, \quad D_{\mathrm{err}} < q^2, \quad \left|\frac{c}{k} - Y\right| < \frac{1}{2}, \quad \operatorname{sgn}(c) = \operatorname{sgn}(kY). \tag{25}$$

Second, the strong and auxiliary Pell equations, together with $c^2 \mid I$, recover the missing main Pell ordinate by absolute index folding. Third, the *pre-projection* part of the signed rank argument yields

$$p = R = 2t + 1, \quad t \geq 24, \quad |Y| > \frac{(|X| - 3)^t}{2} - 1, \quad |H| > |X|(|X| - 3)^t. \tag{26}$$

It does not assume the main scalar projection. Fourth, after that projection has been recovered, the source orientation and half-binomial arguments regenerate a valid old native certificate and hence a finite syntactic refutation. Using the fourth interface before proving the projection would be circular; this construction does not do so.

For completeness, a canonical source witness from a refutation supplies $a + 2 > 1$ even, $R \equiv 3 \pmod 4$, $c \equiv R \pmod 4$, a positive centered budget, and the old main projection. The fixed-pair interpolation theorems described below apply to these canonical witnesses. These retained source facts are substantial inherited mathematics, rather than consequences of finite polynomial testing.

3.3 Bootstrap without a hidden nonzero assumption

The literal coefficient tuple gives, for every integer assignment,

$$q \equiv 0 \pmod 2, \quad q \equiv 2 \pmod 3, \quad |q| \geq \beta - 1, \quad |X| \geq 4|q|, \quad |Y| \geq |q|^3.$$

In particular H is odd and nonzero, and $\Delta > 1$ with $\Delta \equiv 3 \pmod 4$. Writing $\ell = XY^2$ gives $D_t = 1 + \ell(\ell + 1)k^2 \geq 1$. It is odd, and it is one modulo three because ℓ is zero or two modulo three.

Lemma 3.1 (The reduced guard)**.** *The budget $\mathcal{B}$ is divisible by four for every integer assignment. At a zero of (23), one has $S \neq 0$, $srcI \neq 0$, $\mathcal{B} \geq 2$, $D_g > 0$, and $D_a > 0$.*

Proof. The integers q, A_0, B_0 are even. Therefore both q^2 and D_{err} are divisible by four, as is the complete expression for $\mathcal{B}$ in (17). It follows that D is divisible by four and $D_g \equiv 1 \pmod 4$. This makes the old external factor four in $1 + 4DD_ts^2$ unnecessary.

If $S = 0$, then $u = b = e = 0$, and the polynomial is A_c^8. Let $\pi(x) = (x^2 + 3)(x^3 - 19)$. Since

$$A_c = H^5\pi(T + K/H),$$

it cannot vanish: the quadratic factor has no real root and the cubic has no rational root. Thus $S \neq 0$, so $s, r \neq 0$.

At this point $c \neq 0$ must not yet be assumed. If $c = 0$, then $I = Q = 0$, $D_f = 1$, $\delta = C_y = -1$, and $D_a = 1$. If $c \neq 0$, then $|I| \geq 2c^2$ and $D_f > |c| + 1$. In the integer expression $V = D_f(c - R + 4IT) - c$,

either the parenthesized integer is zero, giving $|V| = |c| \geq 1$, or $|V| \geq D_f - |c| > 1$. Hence $Q > 1$, $C_y > 0$, and $D_a > 0$. Thus D_a is positive in both cases.

If $D_g < 0$, its radical in (24) is the sole nonzero imaginary term: D_f, D_a are positive, $D_t \geq 1$, and $S \neq 0$. No conjugate factor can vanish. Since $D_g \equiv 1 \pmod 4$, it follows that $D_g > 0$. If $\mathcal{B} \leq 1$, its divisibility by four gives $\mathcal{B} \leq 0$, so $D \leq -4$ and $D_g \leq -3$, a contradiction. Consequently $\mathcal{B} \geq 2$.

The disk argument now proves (25). Because Y is a nonzero integer, that strip excludes $c = 0$. Thus $I \neq 0$ as well. □

3.4 Separating three radicals and recovering the main root

At a zero, divide the vanishing conjugate factor by S. After a choice of signs, the rational number on its left is a signed sum

$$2\sqrt{D_f} + \sqrt{D_a} + 4\sqrt{P}. \tag{27}$$

All three radicands are positive odd integers. Indeed I is even, so $D_f \equiv D_a \equiv 1 \pmod 4$, while D_t, D_g are odd. For each odd integer radicand, factor its square root as an odd integer times the square root of its odd squarefree part. In any nontrivial squareclass occurring in (27), the coefficients have distinct 2-adic valuations drawn from $1, 0, 2$. The term of least valuation cannot cancel with the others. Linear independence of distinct squarefree radicals then forces all three roots to be rational, hence integral.

The identity

$$\gcd(D_t, D_g) = \gcd(D_t, 1 + DD_t s^2) = 1 \tag{28}$$

shows that the positive square $P = D_t D_g$ has two separately square factors. Thus there exist integers $f, \tau, g, \widetilde{y}$ with

$$f^2 = D_f, \quad \tau^2 = D_t, \quad g^2 = D_g, \quad \widetilde{y}^2 = D_a.$$

Put $y = \widetilde{y}/\delta$ initially in $\mathbb{Q}$. It satisfies $QV^2 - (Q-1)y^2 = 1$. The retained adjacent-coefficient Pell descent makes y an integer. Its hypotheses $Q > 1$, integral V, and square $Q = \Delta^2 I^2$ have already been established. The signed factor now also gives $S \mid A_c$.

For clarity, the main-root recovery occurs before scalar projection. Set $B = |a+2|$, $C_* = |c|$, and $I_* = |I|$. Strong Pell classification writes

$$|f| = X_B(m_0), \qquad I_* = Y_B(m_0), \qquad |f| > 2I_*.$$

Here $X_B(j), Y_B(j)$ are the usual integer coordinates of $(B + \sqrt{B^2-1})^j$. Classification of the auxiliary norm and its two residues gives an odd index ν and $Y_B(\nu) \equiv \pm C_* \pmod{|f|}$. Fold that index modulo $2m_0$ and reflect to $0 \leq p \leq m_0$. Since $Y_B(p) + C_* \leq 2I_* < |f|$, no modular wrap is possible, and $C_* = Y_B(p)$ with $p \geq 1$.

The two residues used here are unchanged:

$$V \equiv -c \pmod f, \qquad V \equiv -R \pmod c.$$

The second follows directly from $V + R = \Delta I^2(c - R) + 4ID_f T$ and $c^2 \mid I$. Lucas composition supplies the required rank divisibility with an even multiplier I/c^2; no odd-multiplier normalization is assumed. The retained pre-projection argument now applies and proves (26).

3.5 Powered divisibility replaces a large carrier

The old construction used $S = IH^5$ to force $H \mid K$. Our smaller $S = srH$ forces only $H \mid K^5$. The exact coefficient-size bounds are strong enough to recover the same final projection from this weaker statement.

Let

$$K_{\mathrm{quad}} = X(X + 2ac) - 1, \qquad \sigma = \mathrm{sgn}(c), \qquad \Pi = (\sigma X - 2^R)(2^R \sigma X + 1).$$

The literal identity

$$K = HXc - 2K_{\mathrm{quad}} \tag{29}$$

is verified by expansion. Since $H \mid S \mid A_c$ and $A_c \equiv K^5 \pmod{H}$, we obtain $H \mid K^5$. Oddness of H then gives $H \mid K_{\mathrm{quad}}^5$. The signed main Pell projection identity supplied by the recovered main ordinate is

$$2^R K_{\mathrm{quad}} \equiv \Pi \pmod{H}.$$

Raising this congruence to the fifth power gives $H \mid \Pi^5$. Only powers of two, which are units modulo the odd H, are cancelled.

Lemma 3.2 (The fifth power is smaller than the modulus)**.** *Under the established pre-projection hypotheses,* $|\Pi|^5 < |H|$.

Proof. Put $x = |X|$. The unchanged actual coefficient tuple gives $x \geq 2^{64}$, far more than is needed. With $R = 2t + 1$ and $t \geq 24$, elementary estimates give

$$|\Pi|^5 < 2^{20t+20} x^{10}, \qquad |H| > x(x-3)^t \geq \frac{x^{t+1}}{2^t}.$$

The first inequality follows by bounding each of the two factors in Π by $2^{R+1}x$. The comparison needed for the two displayed bounds is $x^{t-9} > 2^{21t+20}$. It follows from

$$64(t-9) - (21t + 20) = 43t - 596 \geq 436 > 0.$$

□

Divisibility and the strict height bound force $\Pi = 0$. The second factor of Π is odd and therefore nonzero, so $\sigma X = 2^R$. The retained orientation and half-binomial proof can now be used: it gives $X, c, k, Y > 0$, the required strict native strip, and a fresh valid old certificate. This completes soundness of the new equation relative to the inherited native calibration.

3.6 Completeness with a fixed strong pair

Begin with a canonical outer witness supplied by a finite syntactic refutation in the retained source theorem. Its $\mathcal{B}$ is positive, $D_t = \tau^2$, and DD_t is a positive nonsquare: it is two modulo three. Ordinary Pell existence supplies positive g, s such that

$$g^2 - DD_t s^2 = 1.$$

This directly gives the reduced guard $D_g = g^2$. There is no need to square this unit to create an extra factor four.

Choose one positive strong pair I, f with $4sc^2 \mid I$ and put $r = I/(2c^2)$. Then $r > 0$ and $2s \mid r$. Every prime divisor of $M = sr$ divides I. Hold this strong pair, r, and M fixed.

The supported-modulus interpolation theorem of [2] says that every residue class of T modulo any such M is realized by an integer auxiliary Pell witness y for this fixed strong pair. It applies to the same canonical even base and the same auxiliary chart used here. Its proof does not depend on the choice of determinant or centered-budget guard. The independent review checks the hypotheses $a + 2$ even, $R \equiv c \equiv 3 \pmod 4$, and $4sc^2 \mid I$ explicitly.

We recall the mechanism because the order of quantifiers matters. Writing $I = Y_A(m_0)$ and $f = X_A(m_0)$, the exact period needed modulo f^2 is $\lambda = 4m_0 f$. Set $d = \operatorname{lcm}(\lambda, 4I)$ and vary a starting auxiliary index through $\nu(h) = \nu_0 + dh$. The auxiliary coordinate

$$T(h) = \frac{X_{\Delta I}(\nu(h))/(\Delta I) + c - f^2(c - R)}{4If^2} \tag{30}$$

is an integer. Its denominator is part of the interpolation proof, not part of the final polynomial. For every prime $p \mid I$, Pell addition and valuation lifting give

$$v_p(T(h_1) - T(h_2)) = v_p(h_1 - h_2). \tag{31}$$

The exact rank valuations give $v_p(d) = v_p(4I)$; the factor f in λ cannot be omitted. For $p = 2$, the first-coordinate difference has valuation $2(v_2(\Delta I) + v_2(d(h_1 - h_2))) - 1$, strictly larger than the ordinate contribution because the index difference is divisible by four. Thus the two-adic case is included separately. Equation (31) induces a permutation of each supported prime-power residue ring. Chinese remaindering of the corresponding h classes proves the full supported-modulus statement.

The polynomial $\pi(x) = (x^2 + 3)(x^3 - 19)$ has a root modulo every positive integer. The inherited elementary proof treats the exceptional primes $2, 3, 19$, uses lifting at the other primes, and combines the roots by CRT. Since the canonical projection gives $H \mid K$, write $\kappa = K/H$. Choose a residue with

$$\pi(T + \kappa) \equiv 0 \pmod{M},$$

and only then choose its interpolated auxiliary witness. It follows that

$$\frac{A_c}{S} = \frac{H^4 \pi(T + \kappa)}{sr} \in \mathbb{Z}.$$

Finally set

$$n = \frac{A_c}{S} + 2f + \delta y + 4\tau g - QV. \tag{32}$$

One conjugate factor in (24) now vanishes, giving an integer zero of $\mathcal{E}_{464}$.

The witness order is outer native data, guard Pell unit, one strong pair, carrier residue, auxiliary T, y, and finally n. The strong pair is never changed after the residue is prescribed. The new construction preserves existence of zeros; it does not identify old and new auxiliary tuples pointwise.

3.7 Exact degree and paid arithmetic cost

The principal degree ledger is

Expression	I	S	D_f	Q	V	D_a	P	A_c
Degree	3	8	18	30	23	106	94	35
Expression	u	b	e	$Z_0 = SQV$	$Z_1 = Sn - A_c$			
Degree	34	122	110	61	35			

The center uses the already computed QV. Its exact cancellation is

$$b - Z_0^2 = S^2(-QV^2 - Q + 1). \tag{33}$$

Thus $\omega = (Z_0 + Z_1)^2 - b$ has degree at most 96. Rewriting the identical polynomial as

$$\left[(\omega - u - e)^2 - 4(ue + b(u + e))\right]^2 - 64ube(Z_0 + Z_1)^2 \tag{34}$$

shows that the unique highest-degree part inside the outer square is $-4b_{122}e_{110}$, of degree 232. The last product has degree $34 + 122 + 110 + 122 = 388$. Therefore the degree is exactly 464, with leading form $16b_{122}^2e_{110}^2$.

The independently checked explicit coefficient

$$[j^{260}w^{56}v^{64}C^4s^{12}r^{24}c^{32}h^{12}]\mathcal{E}_{464} = 2^{484}3^{318}\beta^{260} > 0 \tag{35}$$

proves attainment for the actual fixed tuple. The exponent sum is 464. For the direct companion, Z has degree 35 and the leading part is b_{122}^4, of degree 488. An attained coefficient is

$$[j^{248}w^{64}v^{56}C^8s^8r^{40}c^{64}]\mathcal{E}_{488} = 2^{480}3^{312}\beta^{248} > 0.$$

The degree certificate uses exact leading homogeneous arithmetic and the cancellation identity, not a numerical evaluation of the polynomial at a few points.

The delivered graphs count every live arithmetic node, authenticate all six coefficient payloads, and reconstruct the coefficient recipe. The 122-operation count is separate from the cancellation-aware degree argument: an uncollected syntactic degree bound for the shortest graph would overestimate the true degree.

Theorem 3.3 (The eleven-integer equation)**.** *For the six fixed coefficients supplied with the packet, $\mathcal{E}_{464}$ has eleven unrestricted integer unknowns, exact degree 464, and a 122-operation division-free evaluation circuit. The retained source calibration and the arguments above prove*

$$\mathsf{ZFC} \vdash [\forall x \in \mathbb{Z}^{11}\ \mathcal{E}_{464}(x) \neq 0] \longleftrightarrow \mathrm{Con}(\mathsf{ZFC}).$$

The same conclusion holds for the degree-488, 121-operation direct companion.

4 Calibration to syntactic consistency

It is useful to separate the two sources because they certify the same logical statement through different arithmetic representations.

4.1 The direct C222 source for the height-one quartic

The supplied `height_one/source/equation.json` fixes 519 integer variables and 321 quadratic residuals R_j. It is copied byte for byte from the centered direct endpoint in [2]. Its SHA-256 digest is

`2a7dec9de97fdf80de1fdef4977b6cdf096bff5fc1dc7a586cdb5fede5333016`.

The inherited ordinary calibration is

$$\mathsf{ZFC} \vdash \left[\exists x \in \mathbb{Z}^{519} \bigwedge_{j=1}^{321} R_j(x) = 0\right] \longleftrightarrow [\text{there is a finite C222 refutation}] \longleftrightarrow \neg\,\mathrm{Con}(\mathsf{ZFC}). \tag{36}$$

The source uses a finite numerical proof calculus, a direct arithmetic encoding of its certificates, Pell and CRT arithmetic, integer sign representations, exact monic eliminations, and an integral translation. It does not use a simulation of K215 or a specialization of Jones's universal polynomial.

The source logical endpoints are

`KEndpoint.c222_iff_zf_refutation`
`ConstructibleChoice.zf_refutation_iff_zfc_refutation`.

These are inherited source interfaces. The direct arithmetic bridge between the literal residuals and those endpoints is an ordinary proof, not a theorem newly checked by Lean in this packet. The source's main ordinary proof excerpt is retained in `height_one/source/section.tex`; additional arithmetic and calculus proof dependencies are retained in `height_one/source/reading/`, with a reading order and exact source hashes. This curated supplement is not a full ancestor compiler replay or a transitive Lean dependency closure. In particular, the signed comparison index theorem is an ordinary extension of the natural-coordinate Pell criterion; the natural-coordinate theorem alone does not cover arbitrary signed comparison coordinates.

The new coefficient compiler has an unconditional algebraic theorem relative to this literal input system. It proves an explicit bijection of zero sets and the exact value identity (12). Consequently it does not require another assumption about proof coding, Pell completeness, or the consistency of any theory. Composing its finite proof with (36) yields (14).

4.2 The native source for the eleven-variable equation

The eleven-variable construction inherits the coherent source34 coefficient tuple and the ordinary native/Pell/refutation calibration retained by [3]. The coefficient recipe fixes all six integers by finite computations and payload hashes. It neither asks whether ZFC is consistent nor searches for a refutation to choose a coefficient. Its semantic route includes the earlier native encoding and K215 calibration; it is distinct from the direct C222 route above.

The new proof establishes both reductions needed to reuse that calibration. Starting from a zero, it proves nonzero conditions, positivity, radical integrality, the main Pell ordinate, rank and ratio bounds, the exact projection, and then reconstructs a valid old source witness. Starting from a source witness, it chooses a new guard unit, a fixed strong pair, an interpolated auxiliary residue, and the final integer n. It is an equivalence of zero existence with regenerated auxiliary witnesses. An equality of old and new polynomials is not asserted.

4.3 What is proved inside ZFC

Each finite polynomial is an explicitly specified object of integer arithmetic. ZFC proves the required facts about ordered integers, finite sequences, elementary Pell theory, divisibility, and the finite proof transformations used by the retained sources. Thus the combined ordinary arguments prove the equivalences (1) for the stated polynomials. This formalizability claim is different from providing an encoded derivation in a particular implementation of ZFC or a transitive Lean/Rocq build.

No consistency hypothesis is needed to state or prove the reductions. In the usual metamathematical reading, if ZFC is consistent then the equations have no integer zeros. The report does not prove that premise or the resulting nonvanishing statement outright. Equally, an integer zero would, through the specified reductions, encode a finite syntactic refutation. Numerical experiments cannot distinguish these alternatives.

4.4 Priority and attribution

The height-one block-and-cycle compiler was derived in this round and reviewed independently within the AI-assisted workflow. Its elementary energy identities are not claimed as new identities in mathematics. The concrete coefficient-height improvement is measured against the incoming

height-two artifact, not against an exhaustive historical classification of restricted-coefficient Diophantine equations.

The degree-464 equation combines the incoming coprime projection and powered saturation/interpolation results, with the explicitly stated guard and center refinements. These earlier mathematical contributions remain credited. The provided independent reviews are separate agent checks in this workflow; they are not external human endorsements or formal-kernel certificates. Inherited notices and licenses are retained.

5 Explicit artifacts and reproducibility

The archive is intended to be integrated as one research packet. The new checker entry points use relative paths and can be run offline. The complete polynomial data are supplied, rather than only instructions to select a polynomial after a future computation.

5.1 The height-one polynomial

`height_one/output/polynomial.json` contains the ordered coordinate metadata and every one of the 34,086 nonzero coefficients. A sparse term $[c, [i_1, \ldots, i_k]]$ denotes $cz_{i_1} \cdots z_{i_k}$, with zero-based coordinate indices and repetition denoting powers. The empty index list denotes a constant. The companion `polynomial.txt` prints the entire ordinary equation in the same coordinate order.

Metric	Count	Degree of monomial	Count
Original coordinates	519	0	1
Original quadratic rows	321	1	2
Binary constant wires	96	2	20,853
Scaled-factor wires	155	3	2,378
Normalized coordinates	770	4	10,852
Normalized rows	572	Total	34,086
Signed term occurrences	2,378	Coefficient +1	21,934
Equality-cycle dummies	297	Coefficient −1	12,152
Final coordinates	7,431	Unit aliases	1,208

All 519 input coordinates, all 770 normalized coordinates, and all 7,431 output coordinates occur. The exact dimension identity is $3 \cdot 2378 + 297 = 7431$. There are 52 singleton alias groups; they require no agreement cycle. The unit group already has more than two members and needs no padding dummy.

The complete correspondence certificate records every constant and scaling definition, signed occurrence, alias pair, product witness, agreement cycle, and unit-path index. It is sufficient to reconstruct the lifting and projection maps without importing the original proof compiler. The independent verifier imports neither the new producer nor upstream arithmetic code. It checks all 321 source-row pullbacks as exact polynomial identities and independently expands the full $2\mathcal{H} = \sum Q_i^2$ certificate.

The release also checks total and individual degrees, the coefficient alphabet, constant term, variable occurrence, and exact equality of the JSON and plaintext polynomials. Two actual modified-artifact checks flip a coefficient and remove the unit-anchor constant; both are rejected by the full verifier at the expanded-polynomial comparison. Five deterministic signed evaluations supplement the exact checks; they are not the proof of a universal identity.

5.2 The eleven-variable circuits

The two circuit JSON files under `eleven/` list integer leaves, eleven coordinate leaves, six authenticated fixed-coefficient leaves, and binary arithmetic nodes. Their output register fixes the complete polynomial. `eleven/coefficients/` contains all six compressed integer payloads, byte-order and binding information, and the finite source recipe. A coefficient leaf therefore denotes a literal fixed integer, not an unbound parameter.

The independent checker verifies the universal three-radical norm identity, the actual graph definitions, dependency order, liveness, variable set, arithmetic count, exact degree cancellations, and attained leading coefficients. It authenticates the whole coefficient tuple and reconstructs the native recipe. The proof review separately checks the unbounded logical implications, including the case $c = 0$, the reduced guard, the exact Pell period, the supported-modulus hypothesis, and the order of witness selection.

5.3 Running the checks

From the extracted packet root, the supported release command is

```
python3verify_all.py
```

It runs the two independent verifiers in a temporary copy, compares deterministic regeneration with the shipped data, and checks the release manifest. Both new compilers and their exact verifiers use only the Python standard library; no SymPy, theorem prover, or network access is required. The release runs ordinary and optimized Python checks. The new verifiers use explicit exceptions and authenticate the rebuilt coefficient values in both modes, even when assertions in an inherited recipe are disabled.

To regenerate the PDF, run two passes, or use `latexmk`, from `report/`. The LaTeX source and all included sections are supplied. PDF bytes may differ with TeX versions, metadata, or timestamps; polynomial JSON regeneration is byte-for-byte deterministic.

The included receipts record actual successful executions. They certify the finite artifact checks described above. They do not constitute a satisfiability search, a proof of Con(ZFC), or an automatically generated derivation of the inherited unbounded arithmetic calibration.

6 Further research questions

6.1 Reduce dimension while retaining height one

The general compiler deliberately assigns two fresh aliases and one product witness to every signed occurrence. This gives a transparent collision proof but is not a minimal allocation. Several precise optimization problems follow. Which linear occurrences can remain unpadded while preserving disjointness between the quadratic agreement terms and the row-block terms? Can two occurrences share a product witness without introducing another zero or a coefficient of magnitude two? Which agreement classes can be represented by a smaller positive semidefinite form with the same integer kernel and coefficient alphabet? Any optimization should supply both its zero-set proof and a complete collected-coefficient audit.

The current binary normalization uses 96 constant wires and 155 shared scaled-factor wires. Alternative addition chains, signed-digit constant representations, and selective use of a different 519- or 517-variable source may reduce the normalized occurrence count before the height-one compiler is applied. The correct objective is the final count $3L + d_2 + \delta_u$, not the smallest intermediate source dimension alone.

6.2 Compile the low-variable circuit directly

Another route is to quadratize the 122-operation circuit and then apply the height-one compiler. Its native coefficients have very large bit lengths, so explicit coefficient construction may dominate the resulting variable count. A meaningful comparison must include those constant wires. Treating the six source ports as free constants in a circuit count and then omitting their cost in a height-one encoding would not be a fair comparison.

6.3 Lower the eleven-variable degree further

The degree-464 leading form is $16b_{122}^2e_{110}^2$. Thus the fused guard radical and the auxiliary radical are the present bottleneck. Reducing only an expression that lies below those two leading degrees will not lower the final degree. Possible targets include a smaller budget guard preserving the unconditional parity information, a different coprime fusion, or a carrier that continues to exclude both $r = 0$ and $s = 0$ with less degree.

The supported-modulus theorem must continue to be applied with a fixed strong pair. A proposed carrier change is incomplete if it chooses a residue first and then silently changes that pair to realize the residue. Likewise, a weakened projection must compare the entire powered divisibility product with the actual modulus. The generous authentic coefficient bound may permit other exponents or smaller scales, but each such claim needs a new quantitative proof.

6.4 Formalize the local new transformations first

The height-one compiler offers a relatively compact formalization target: finite sparse polynomial identities, a graph-cycle kernel, nonnegativity, and an explicit zero-set bijection. A proof assistant could first verify the generic compiler and then check the literal occurrence certificate. This would avoid formalizing C222 arithmetic again merely to validate the coefficient transformation.

For the eleven-variable equation, useful separable targets are the guard divisibility lemma, three-root separation with weights $2, 1, 4$, the powered projection bound, and the supported-modulus interpolation interface. The latter needs explicit attention to the exact period $4m_0f$ and the two-adic valuation step. An eventual end-to-end result must connect these local proofs to the concrete source equation and its internal consistency statement, preserving the exact coefficient tuple throughout.

6.5 Define a reproducible simplicity objective

The two new endpoints make the tradeoff visible. Height one is attained with thousands of variables; eleven variables are attained with large fixed coefficients and higher degree. A next comparison could minimize a specified encoded byte length, weighted arithmetic cost, or lexicographic tuple of degree, dimension, height, and support. Without such a choice, a global claim of "the simplest equation" is not a well-defined conclusion of this packet. What is established here is two concrete, fully specified improvements and the proofs needed to evaluate them.

References

[1] Vladimir Reshetnikov and ProveIt contributors, *ProveIt: Diophantine consistency research*, snapshot `45b399ec51dfbecb891dd51d8a9e4b6b7c361ac6`, 24 October 2026. Immutable repository tree.

[2] ProveIt incoming research packet, *Diophantine consistency equations with powered saturation and factorial carriers, Round 14*, 24 October 2026. Repository archive `ConZFC_Powered_Saturation_Factorial_Carriers_Round14_2026-10-24.zip`; the source tuple, direct 519-variable system, and selected ordinary proof dependencies are identified in the supplied provenance files.

[3] ProveIt incoming research packet, *Diophantine consistency equations by coprime projection*, 24 October 2026. Repository archive `ZFC_Diophantine_Coprime_Projection_2026-10-24.zip`, particularly `integer/PROOF.md` and its retained centered source proofs.

[4] ProveIt incoming research packet, *Signed powers and coefficient height two*, 24 October 2026. Repository archive `ConZFC_Signed_Powers_Height_Two_2026-10-24.zip`, `height_two/PROOF.md` and the `anchored/` endpoint.

[5] J. Bayer, M. David, M. Haßler, Y. Matiyasevich, and D. Schleicher, *Diophantine equations over $\mathbb{Z}$: universal bounds and parallel formalization*, arXiv:2506.20909v3, 22 October 2026. https://arxiv.org/abs/2506.20909.

[6] Z.-W. Sun, *Ten unknowns for Hilbert's tenth problem over the integers*, arXiv:2609.01594v2, 2 September 2026. https://arxiv.org/abs/2609.01594.